

Epic 2: Visualizing and Analysing the Data

Description

This phase focuses on exploring and understanding the Credit Card Approval Prediction dataset through comprehensive data visualization and statistical analysis. The objective is to gain meaningful insights into applicant information, financial attributes, and credit-related features before building machine learning models. Exploratory Data Analysis (EDA) is performed to identify data distributions, trends, correlations, missing values, outliers, and feature relationships that may influence credit card approval decisions.

The dataset is analyzed using Python libraries such as Pandas, Matplotlib, and Seaborn to generate visual representations including count plots, histograms, box plots, correlation heatmaps, and distribution graphs. These visualizations help understand applicant characteristics such as income level, employment status, education level, family status, housing type, and credit history.

Statistical analysis is also performed to summarize the dataset using measures such as mean, median, standard deviation, minimum and maximum values, and percentiles. Univariate analysis is used to study individual features, while multivariate analysis identifies relationships between multiple variables that impact approval prediction. The insights obtained during this phase improve feature engineering, data preprocessing, model selection, and overall prediction accuracy.

Activities Performed

- Importing the required Python libraries.
- Reading and exploring the dataset.
- Performing univariate analysis.
- Performing multivariate analysis.
- Generating descriptive statistical summaries.
- Visualizing important feature relationships.
- Identifying trends, correlations, and anomalies.

Outcome

After completing this phase, the dataset is thoroughly explored and validated for machine learning. The analysis provides a clear understanding of data quality, feature importance, and applicant behavior, enabling the development of a more accurate and reliable Credit Card Approval Prediction model.